EXERCISES IN STRAIN ANALYSIS

The following exercises are designed to take you through some simple strain analysis techniques using situations that you might well encounter while mapping. The first focuses on recovering the 3d state of strain from measurements of 2d strain on three differently oriented planes; the other three explore widely used techniques to determine 2d strain.

Exercise 1

In this exercise we seek to determine strain on three orthogonal 2d sections, and then to represent the 3d state of strain obtained from these measurements (a) on a Flinn diagram and (b) on a Hsü diagram. The key parameter needed for the 3d representations is the ellipticity of the strain ellipse on the three principal planes of the strain ellipsoid (that is, the planes containing the X and Y directions, the X and Z directions, and the Y and Z directions, where X, Y, and Z are the long, intermediate, and short axes of the strain ellipsoid). These ellipticities are given by

$$R_{XY} = \frac{2(1+e_1)}{2(1+e_2)} = \frac{(1+e_1)}{(1+e_2)} \quad ; \quad R_{XZ} = \frac{(1+e_1)}{(1+e_3)} \quad ; \quad R_{YZ} = \frac{(1+e_2)}{(1+e_3)}$$

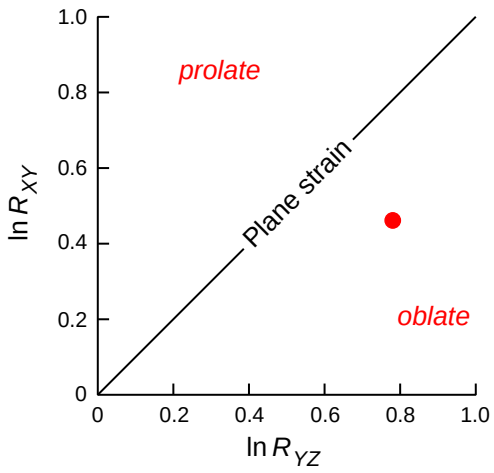
Note that if one knows any two of these ellipticities, then the third may be obtained from them, for example,

$$R_{YZ} = \frac{(1+e_2)}{(1+e_3)} = \frac{(1+e_1)}{R_{XY}(1+e_3)} = \frac{R_{XZ}(1+e_3)}{R_{XY}(1+e_3)} = \frac{R_{XZ}}{R_{XY}}$$

Figure 1 shows photographs of some deformed lapilli on three orthogonal sections through a deformed Ordovician tuff from near Coniston. Assuming that the three sections correspond to the three principal planes of strain and that the lapilli were originally spherical, show the state of strain (a) on a Flinn diagram and (b) on a Hsü diagram.

Method

1. On each photograph measure the long and short axes of 10 ellipses. Select ellipses with a range of different sizes. Determine the aspect ratio for each measured ellipse.
2. For each photograph determine the mean ellipse aspect ratio. The three photographs correspond to the XY, XZ and YZ planes − which is which? Top: XZ (X horizontal); Middle: YZ (Y horizontal); Bottom: XY (X horizontal)
3. Note that the mean ellipse aspect ratio on the XY plane is R_{XY}, and mean ellipse aspect ratio on the YZ plane is R_{YZ}. Hence take natural logarithm of the mean aspect ratio of the ellipse on each of the relevant two planes and plot the result on the Flinn diagram below. Is the ellipsoid in the prolate or oblate field?



R_{XZ} = 4.06
R_{YZ} = 2.18
R_{XY} = 1.59

4. Hsü's diagram is plotted in terms of logarithmic strains, ε_1, ε_2, and ε_3. We can work with the mean aspect ratios, R_{XY}, R_{XZ}, and R_{YZ} because, for example,

$$R_{XY} = \frac{2(1+e_1)}{2(1+e_2)} = \frac{(1+e_1)}{(1+e_2)}$$

and so

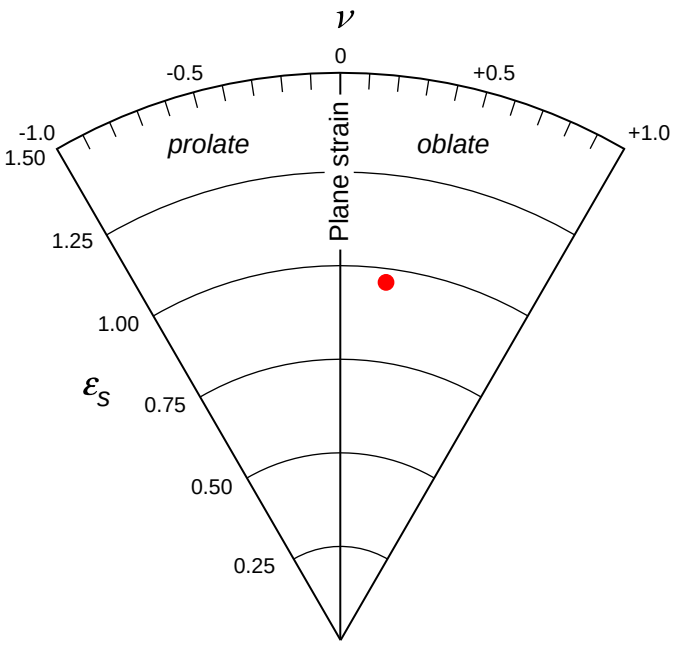
$$\ln R_{XY} = \ln\frac{(1+e_1)}{(1+e_2)} = \ln(1+e_1) - \ln(1+e_2) = \varepsilon_1 - \varepsilon_2$$

Hence it is trivial to show that

$$\nu = \frac{2\varepsilon_2 - \varepsilon_1 - \varepsilon_3}{\varepsilon_1 - \varepsilon_3} = \frac{\ln R_{YZ} - \ln R_{XY}}{\ln R_{XZ}}$$

$$\varepsilon_s = \frac{1}{\sqrt{3}}\left[(\varepsilon_1-\varepsilon_2)^2 + (\varepsilon_2-\varepsilon_3)^2 + (\varepsilon_3-\varepsilon_1)^2\right]^{\frac{1}{2}} = \frac{1}{\sqrt{3}}\left[(\ln R_{XY})^2 + (\ln R_{YZ})^2 + (\ln R_{XZ})^2\right]^{\frac{1}{2}}$$

Calculate ν and ε_s from your mean aspect ratios and plot the result on the Hsü diagram below. Confirm that the ellipsoid is in the same field (oblate or prolate) as identified with the Flinn diagram.



5. Compare the value of R_{YZ} you measured with the value calculated from R_{XZ} and R_{XY}. What are the implications of this comparison? **$(R_{YZ})_{measured}$ = 2.18 ; $(R_{YZ})_{calculated}$ = 2.55. The difference implies either that the section planes are not perfectly parallel to the three principal planes of strain and/or that the lapilli were not all initially perfectly spherical but were in fact ellipsoids with varying orientations. The issue might go away by analysing more lapilli or by employing statistics more effectively (*e.g.*, perhaps the modal value rather than the mean would be better to use if there are some very anomalous shaped lapilli skewing the aspect ratio frequency distribution). The nature of the statistical issue can be examined by looking at the shape of the aspect ratio frequency distribution or by plotting, for each data set, long axis against short axis and seeing the scatter about a linear fit to the data. In advance of doing this, the ellipticity of the strain ellipsoid is not reliably evaluated.**

Exercise 2

In Exercise 2 we examine the usefulness of the Mohr circle construction for the analysis of 2d strain. Remember this utilizes the reciprocal quadratic extension parameter,

$$\lambda' = 1 / (1 + e)^2$$

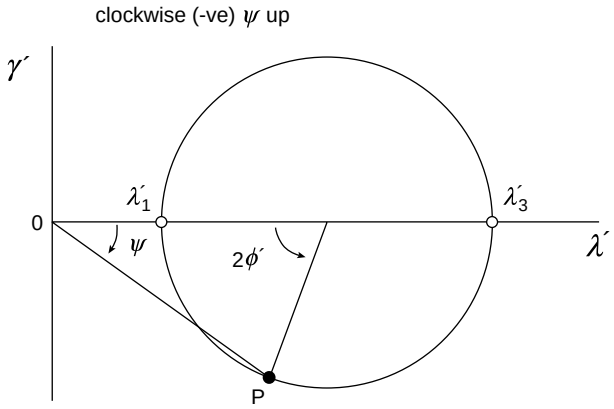
The construction is particularly helpful in the following cases (e_1 is the direction of greatest extension in the plane):

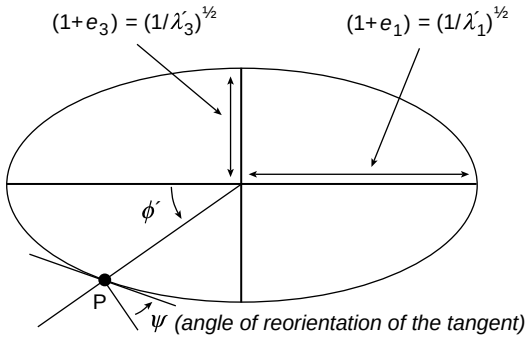
(a) If ψ and ϕ' are known (that is, the angular shear in given direction and the angle that that direction makes with e_1), then the ratio (λ_1' / λ_3') may be determined. If the information is also supplemented by a length change in known direction with e_1, then the actual values of λ_1' and λ_3' may be evaluated and the full strain on the plane can be determined. (See Ragan, p.316-317 for worked example.)

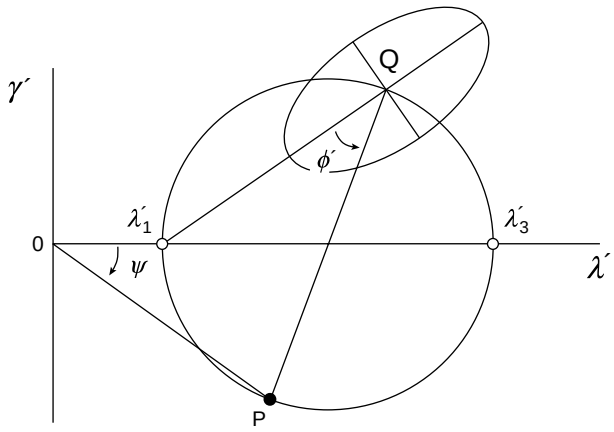
(b) If ψ is known in two different directions and the angle between those directions, then the ratio, (λ_1' / λ_3') may be determined together with the orientation of the e_1 direction. Again, if the information is also supplemented by a length change in known direction with e_1, then the actual values of λ_1' and λ_3' may be evaluated and the full strain on the plane can be determined. (See Ragan, p.318-319 for worked example.)

(c) If length changes are known in two different directions together with the angle each direction makes with e_1, then the full strain on the plane can be determined. (See Ragan, p.325-326 for worked example.)

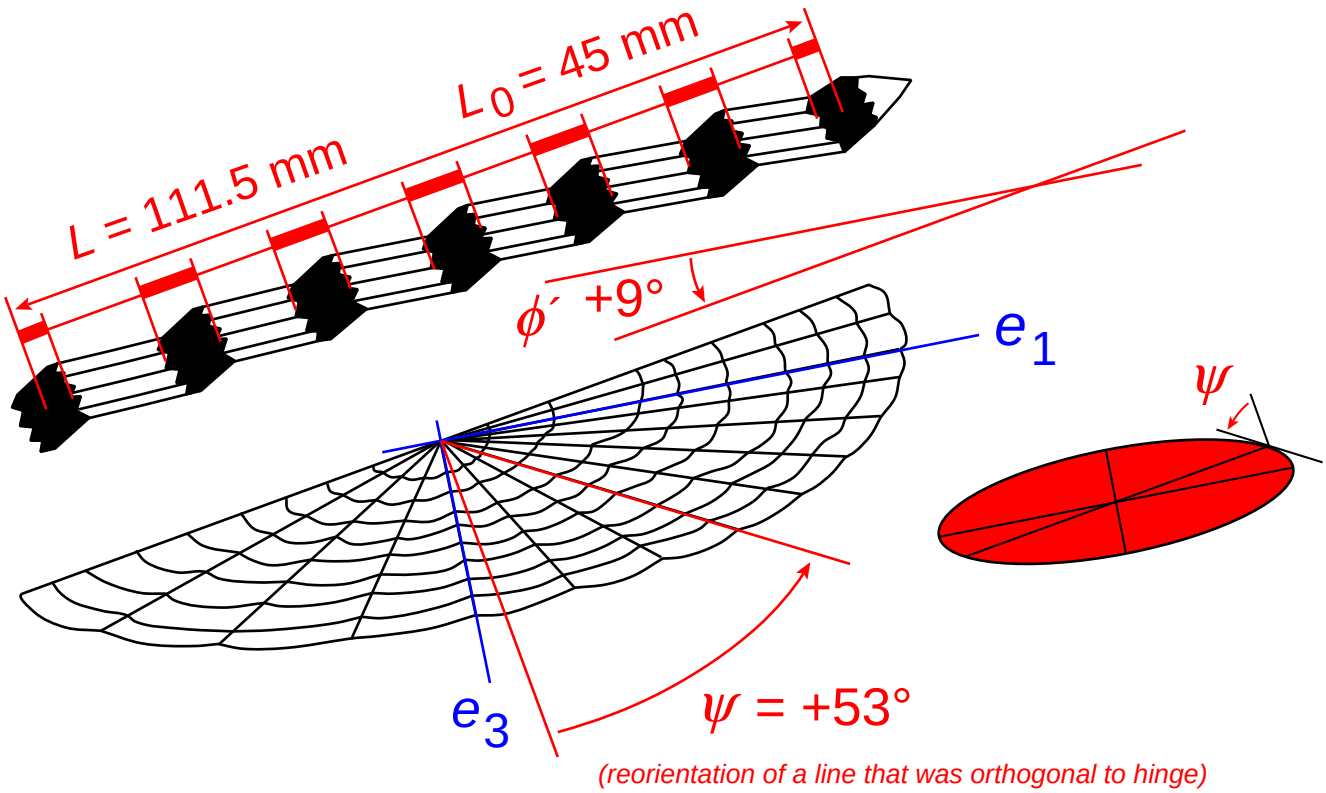
(d) If length changes are known in three different directions then the orientation of e_1 and the full strain on the plane can be determined. (See Ragan, p.327-328 for worked example.)







This following exercise is designed to provide a flavour of the way in which Mohr circle constructions can be used for strain analysis. You are encouraged to look at further examples, many of which involve an imaginative first step, in the texts by Ragan (Chapter 12) and by Ramsay and Huber (Sessions 6 and 8).



The diagram above is a sketch of a block of deformed limestone containing a stretched crinoid stem and a deformed brachiopod. The crinoid stem has calcite fibres between the ossicles; the brachiopod had a rib pattern that was originally radially symmetric.

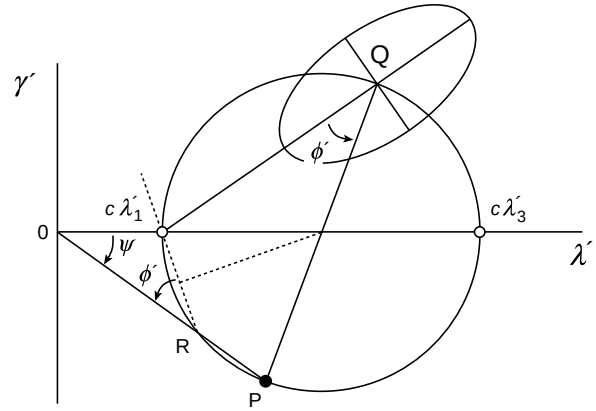
1. Calculate the following strain parameters in the direction of the hinge line of the brachiopod (NB. it is best to measure the final length of the crinoid stem from the middle of the first ossicle to the middle of the last, and in calculating the original length, to use only the half lengths of the first and last ossicle together with the lengths of the five intervening ossicles):

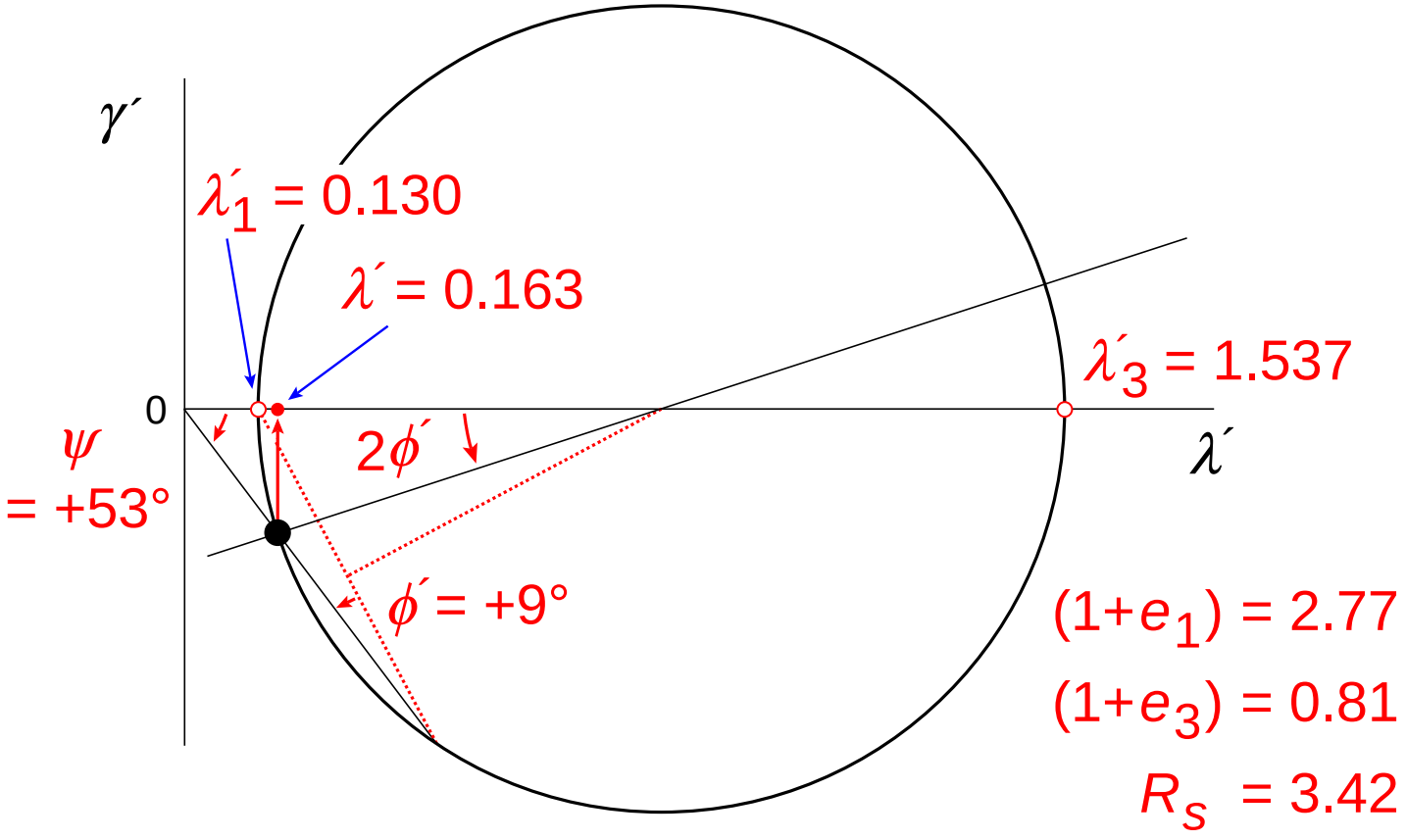
e	$= \Delta L / L_0$	= (111.5 - 45) / 45 = **1.478**
λ	$= (1 + e)^2$	= **6.139**
λ'	$= 1 / \lambda$	= **0.163**
ψ		= **+ 53° (anticlockwise rotation)**
γ	$= \tan \psi$	= **1.327**
γ'	$= \gamma / \lambda$	= **0.216**

2. It seems possible that the crystal fibres between the separated crinoid ossicles are aligned in the direction of maximum extension. Satisfy yourself that the rib pattern on the brachiopod supports this thought, draw on the direction of maximum extension, and determine the angle ϕ' from the maximum extension direction to the hinge line of the brachiopod:

ϕ' = **+ 9° (anticlockwise rotation)**

3. Construct the following Mohr circle diagram:





(a) On a set of λ' γ' axes, draw a line through the origin with a slope angle of ψ (take care to observe the -ve angular shear slopes up / +ve angular shear slopes down convention)

(b) At any convenient distance along the ψ line (R on the diagram above), construct a line at angle ϕ' to it (on the diagram above this is a dashed line). Where this line intercepts the λ' axis is λ_1' times an unknown constant, c.

(c) Draw a line at right angles to this new line starting halfway between R and point $c\lambda_1'$. Where this intercepts the λ' axis is the centre of the Mohr circle.

(d) Calculate the ratio R_s

$$R_s = (1+e_1) / (1+e_3) = (c\lambda_3' / c\lambda_1')^{½} = \mathbf{3.42}$$

(e) From the centre of the Mohr Circle draw a radius at angle $2\phi'$ to the λ' axis (measuring from $c\lambda_1'$). Where it intercepts the circle (P on the diagram above), the value of λ' is that which you obtained in part 1. What therefore are the actual values of the strains?

$(1+e_1)$ = **2.77** $(1+e_3)$ = **0.81**

(f) Draw a strain ellipse to scale and in the proper orientation on the diagram that shows the fossils.

Exercise 3

The Fry method is a simple but very powerful technique for analysing the 2d strain of initially equidimensional objects (*e.g.*, circular clasts, equant grains in a monomineralic rock). Here we apply it to the oolitic ironstone shown in Figure 2 in order to estimate the ellipticity and orientation of the strain ellipse on the thin section plane.

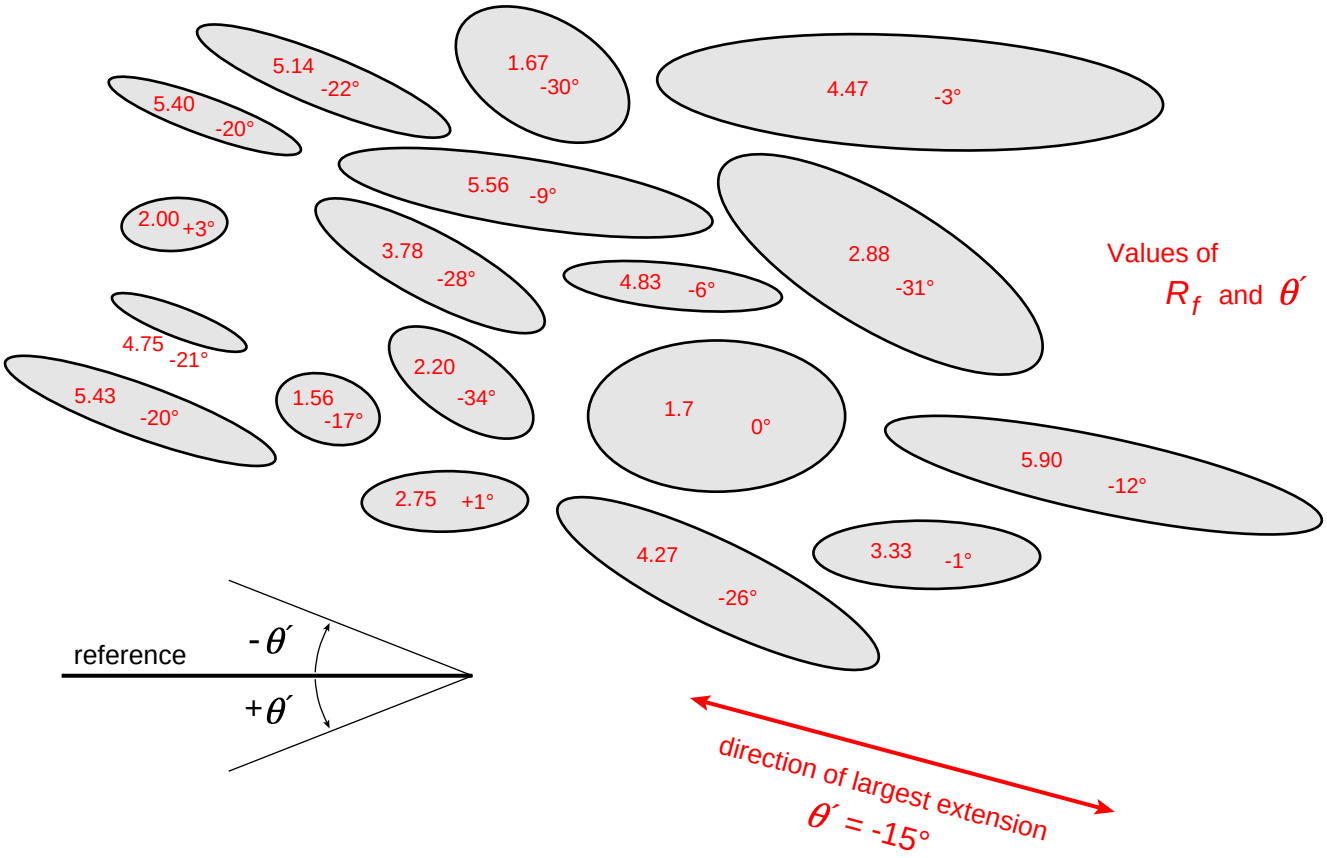
Method

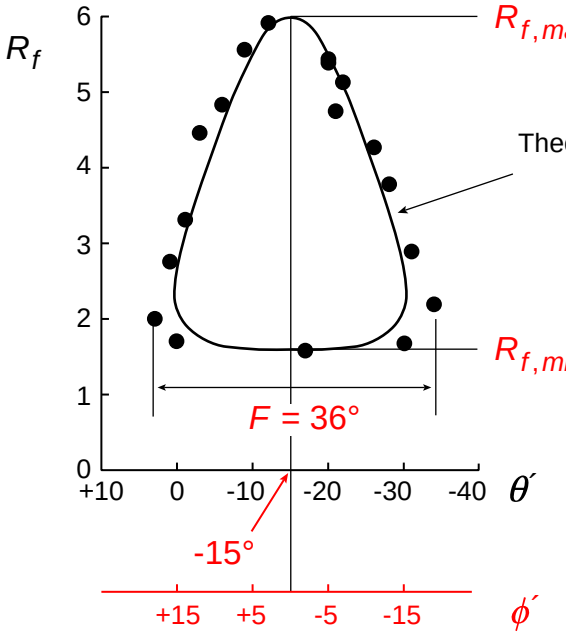
1. On a sheet of tracing paper mark the centres of all the oolites, and number these points. HINT: the exercise will proceed faster if you number the oolites in the centre of the image first and work outwards.
2. On another sheet of tracing paper mark a central reference point. Place this over oolite number 1 and trace the centres of all the other oolites on to the overlay.
3. Move the overlay (without rotating it) so that the overlay reference point lies over oolite number 2, and trace the centres of all the other oolites on to this overlay.
4. Repeat step 3 for all the other oolites. After a few oolites an elliptical point vacancy starts to emerge around the overlay reference point. When it does so, if you want to speed matters up then the grain centres at distances well away from this vacancy need not be marked.
5. The elliptical vacancy has the aspect ratio and orientation of the strain ellipse. Determine the long and short axes of this ellipse by eye and calculate $R_s = (1+e_1) / (1+e_3)$. Also measure the angle between the horizontal and the long axis of the ellipse.

The Fry method as used in this exercise relies on the measured objects being all of about the same size and to be fully dispersed (not clustered). It is easily modified for use with initially equidimensional objects of varying size (see Erslev, 1988, *Journal of Structural Geology* 10:201-209), and formal strategies for fitting the ellipse to the data have also been developed (see Erslev and Ge, 1990, *Journal of Structural Geology* 12:1047-1059).

Exercise 4

The R_f / ϕ' method is widely used to analyse the 2d strain of initially elliptical markers (*e.g.*, clasts in a conglomerate). Here we apply it to the artificial set of deformed elliptical pebbles shown below to illustrate the technique.





$R_{f,max} = 6.00$

Theoretical curve given by the values of R_s and R_i that were obtained from $R_{f,max}$ and $R_{f,min}$

$R_{f,min} = 1.60$

$F = 36°$

$R_s = [R_{f,max}\ R_{f,min}]^{½} = 3.10$

$R_i = [R_{f,max} / R_{f,min}]^{½} = 1.94$

Different means of the R_f data as an approximation of R_s

arithmetic mean = 3.76 ; geometric mean = 3.42 ; harmonic mean = 3.08

Method

1. Just under the elliptical pebbles a horizontal line has been drawn to serve as an arbitrary reference direction relative to which the orientation of the long axes of the ellipses can be measured. Measure the angle θ' that the long axis of each pebble makes with this reference direction, taking care to follow the indicated sign convention (clockwise -ve; anticlockwise +ve).
2. For each pebble, measure the long axis and short axis of the ellipse, and determine the aspect ratio, R_f (long axis ÷ short axis).
3. Plot each pair of (θ', R_f) values as a point on the indicated graph.
4. The points lie on a closed, tear-drop shaped curve, symmetrical about a given value of θ'. That value of θ' is the maximum extension direction. Next to the reference direction, draw a line which is in the orientation of this maximum extension direction. Now relabel the θ' axis on your graph so that it reads in terms of ϕ', the angle measured <u>from</u> the maximum extension direction <u>to</u> the long axis of given ellipse. Again follow the sign convention
5. The maximum and minimum values of R_f lie on the line $\phi' = 0$. What are these values?
6. It is possible to show that the ellipticity of the strain ellipse, R_s, is given by

$$R_s = [R_{f,max} R_{f,min}]^{½}$$

   and that the initial ellipticity of the pebbles, R_i, is given by

$$R_i = [R_{f,max} / R_{f,min}]^{½}$$

   Calculate both R_s and R_i.
7. For given R_i and R_s the shape of the R_f / ϕ' curve is given by

$$\cos 2\phi' = \frac{R_i(R_f^2+1)(R_s^2+1) \pm 2(R_i^2+1)R_s R_f}{R_i(R_f^2-1)(R_s^2-1)}$$

   When applied to this data for the R_i and R_s calculated in Part 6 it is found that the tear-drop shaped curve given by this theoretical formula is a little too narrow (see solution). The maximum width of this curve is referred to as the *fluctuation, F*. The fact that the data has slightly wider fluctuation than that given theoretically suggests that there is a small component of pebble rotation in addition to that produced by the pebble shape change. This could be produced, for example, if the pebbles came in contact with each other or if the matrix and pebbles had significantly different rheological (flow) properties.
8. Because the R_f / ϕ' method is a time-consuming one, attempts have been made to use different types of means of the R_f data as an approximation for R_s. These include

$$\textit{Arithmetic mean} = \frac{1}{n}\sum R_f \quad ; \quad \textit{Geometric mean} = \left(\prod R_f\right)^{1/n} \quad ; \quad \textit{Harmonic mean} = \frac{n}{\sum (1/R_f)}$$

   Calculate these three means. None are expected to give good agreement with the actual R_s at high R_i or at small R_s but the harmonic mean is commonly found to be a good approximation when $R_s > 2$ and $R_i < 2$ are both true.

The R_f / ϕ' method as used in this exercise requires all of the measured objects to have the same initial ellipticity and for the initial orientation of the long axes of these elliptical objects to be random. The fact that in real world situations one or both of these conditions are not generally satisfied complicates the application of the method. Nevertheless, analysis strategies have been developed that make this method a very powerful one (see Lisle, 1985, *Geological Strain Analysis: A Manual for the R_f / ϕ Technique*, Pergamon Press).